

Bruce Nduko

Co-Founder & CTO, Tija Labs · Founder-Engineer · BSc Computer Science

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PROFILE

Founder-engineer building AI-powered productivity tools for young, urban Africans and Kenya's informal economy. Co-founded Tija Labs and lead product and engineering end to end: web and Android apps, payments, infrastructure, security and deployment. Ranked 14th on the Phase 1 public leaderboard of the AWS Trainium Frontier Competition. Completed 22 certifications across AI, software engineering, security and cloud while studying Computer Science full-time.

3 live products	#14 AWS Trainium Frontier (Phase 1)	22 certifications	1,093+ learning hours	3 hackathons & competitions
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EXPERIENCE

Co-Founder & Chief Technology Officer — Tija Labs Kenya · Present

AI product lab building productivity apps for young, urban Africans. "Turn Time Into Value."

- Own product, engineering and infrastructure across the portfolio; co-lead the company with co-founder Jeremiah Rioba (People & Operations).
- Shipped three live products across web and Android — StudySphere, Allama and LeverageX (Google Play).
- Built a complete subscription invoicing and billing system (Node.js/Express, SQLite, Pesapal API v3, PDF invoicing, automated email); selected Pesapal after evaluating gateway availability in Kenya.
- Led pre-deployment security hardening and environment/secrets configuration for production apps.
- Built and deployed tijalabs.com with custom DNS, contact-form email delivery and blog; set up company email, LinkedIn page, WhatsApp Business and brand identity.
- Wrote hiring specs for Backend, Mobile and Frontend engineering roles.

SELECTED PROJECTS

Allama — Productivity Passport Live · Web allama-theta.vercel.app
Turns M-Pesa SMS, job photos and voice notes into verifiable work records for informal-economy workers. Led build, security hardening and production deployment.

StudySphere Live · Web + Android studysphere.it.com
Student productivity platform with an AI assistant (Sage), autopilot study planner, focus timer and exam mode. Deployed on Vercel with custom domain.

LeverageX Live · Google Play [Google Play](https://play.google.com/store/apps/details?id=com.leverage)
Productivity and goal-tracking Android app. Took it through Play Console closed testing to production.

Tija Labs Billing Engine Built · Payments
Subscription invoicing and billing: Node.js/Express, SQLite, Pesapal API v3, pdfkit PDF invoices, nodemailer delivery.

Hali In development · Nairobi alpha
Venue discovery platform with Taste DNA personalization and live venue intelligence (Hali Pulse). Authored a 21-section PRD.

Ledger · A-Inya In development
Ledger: chat-first AI career copilot. A-Inya: nutrition app that helps users decide what to eat.

COMPETITIONS & HACKATHONS

AWS Trainium Frontier Competition — AWS Annapurna Labs 2026
Ranked 14th on the Phase 1 public leaderboard. Co-designed ML models and custom kernels on AWS Trainium2 chips.

Build with Gemini XPRIZE 2026
Submitted 10 original productivity product concepts, including Stakes, Sema Kazi, Alongside, Reflow and Productivity Passport.

Qwen Cloud Hackathon 2026
Produced build specs for Productivity Passport, an SHA Claims Pre-Check Agent and a Skill Verification Agent for Kenya's informal economy.

EDUCATION

BSc Computer Science — Zetech University	Expected 2029
Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Computer Vision, Probability & Statistics, Python, Machine Learning Fundamentals.	
AI Programming with Python Nanodegree — Udacity	In progress
Microsoft AI-901: Azure AI Fundamentals	In progress

CERTIFICATIONS · 22 COMPLETED · 1,093+ HOURS

AWS AI Practitioner Challenge — Udacity	Verified Certificate of Course Completion	Jun 2026
Codecademy		
AI & Data		
Data Scientist: Machine Learning Specialist	90 hrs · Jun 2026	
Data Analyst	70 hrs · Jun 2026	
Deep Learning & Neural Networks	50 hrs · May 2026	
Business Intelligence Analytics	45 hrs · Jul 2026	
Analyze Financial Data with Python	26 hrs · May 2026	
Prompt Engineering Specialist	20 hrs · Apr 2026	
Data Scientist: Natural Language Processing Specialist	Jul 2026	
Security & Cloud		
Cloud Architecture – AWS Specialist	65 hrs · Mar 2026	
Ethical Hacking & Penetration Testing	55 hrs · Feb 2026	
Systems & Linux Administration	40 hrs · Mar 2026	
Cybersecurity Fundamentals	35 hrs · Jan 2026	
Software Engineering		
Full-Stack Engineer		140 hrs · Sep 2026
Front-End Engineer		115 hrs · Aug 2026
Back-End Engineer		105 hrs · Aug 2026
DevOps Engineer		75 hrs · Apr 2026
Languages		
Learn C++		35 hrs · Feb 2026
Learn JavaScript		30 hrs · Jan 2026
Learn Go		30 hrs · May 2026
Learn Python 3		25 hrs · Jan 2026
Learn TypeScript		22 hrs · May 2026
Learn SQL & Relational Databases		20 hrs · Mar 2026

SKILLS

Languages	Python, TypeScript / JavaScript, C++, Go, SQL
AI & Data	LLM-powered features, machine learning, NLP, deep learning, computer vision, data analysis, prompt engineering
Engineering	Node.js, Express, React, REST APIs, SQLite, payments (Pesapal), PDF & email automation, full-stack web
Mobile	Android app publishing and release management (Google Play Console)
Cloud & DevOps	AWS, Vercel, GitHub, CI/CD, DNS & domains, environment & secrets management
Security	Security hardening, penetration-testing fundamentals, Kali Linux, Linux administration
Product	PRDs, product concepting, engineering hiring specs, brand identity, AI-assisted development (Claude, Codex)